Exp. No: 07
Date: 28/9/26.

Sliding Window.

Aim:

To Implement and stimulate the sliding window protocol using C programming to demonstrate frame translation, acknowledgement (ACK), negative Acknowledgment (NACK), and retransmission handling using a circular queue.

Code:

```
#include <stdio.h>
#include <string.h>
#define MAX 100

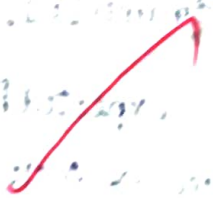
struct Frame {
    int frameNo;
    char data;
};
int main() {
    struct Frame frames[MAX];
    char message[MAX];
    int totalFrames, windowSize;
    int base = 0;
    int expectedFrame = 1
    int lostFrame;
    int lossSimulated = 0;
    int i, end;
    int needRetransmission;
    printf("Enter the message:");
    fgets(message, sizeof(message), stdin);
```

```
printf("Enter Msg");
fgets(message, sizeof(message), stdin);
totalFrame = strlen(message);
if(totalFrame == 0) {
    printf("Msg can't be empty.\n");
    return 1;
}
if ( windowSize > totalFrame){
    windowsize= totalframes;
}
printf("Enter the frame to be lost");
if ( (sscanf("%d", &lostFrame) != 1 ||
    lostFrame < 0 || lostframe > totalFrame) {
    printf("Invalid frame Number:");
    return 1;
}
for (i=0; i < totalframe; i++) {
    frame[i].frameNo = i+1;
    frame[i].data = message[i];
}
printf("Total frame : %d\n", totalframe);
printf("Window Size: %d\n", windowSize);

while ( base < totalframe) {
    end = base + windowSize;
    if( end > totalFrame){
        end = totalframe;
    }
    needTransmission = 0;
```

```
Printf(" Sender Window");
for( i = base; i < end; i++) {
  printf (" [%d:%c]", frames[i].frameNo, frames[i].data);
}
 Printf(" Sender Transmission Frames");
for(i= base; i < end; i++) {
   int frameNo = frames[i].frameNo;
   if (frameNo == lostFrame  && !lossSimulated) {
      Printf(" Frame %d was lost during Transmission;frameNo
       lossSimulated = 1;
       needTranmission = 1;
     Printf("Receiver → NACK %d\n", expectedFrame);
          continue;
   }
Printf(" Receiver ← Frame :%d; Data: %c\n",
      frameNo, frames[i].data);
if (frameNo ==  expectedFrame ) {
   printf("Receiver → ACK %d\n", frame);
    expectedFrame ++;
}
else{
   printf("Receiver discards Out-of-order frame %d \n",
         frameNo);
    printf("Receiver → NACK %d\n", expectedFrame);
       needRetransmission = 1;
    }
}
if( needRetranemission) {
      for(i= expectedFrame - 1; i < end; i++) {
      frame[i].framNo, frames[i].data);
   }
```

```
for (i = expectedFrame - 1; i < end; i++){
    int frameNo = frame[i].frameNo;
printf("Receive");
  if (frameNo == expectedframe) {
      printf(" ACK");
      expectedframe++;
  }
  else {
      printf (" NACK");
  }
}
base = end;
if (base < totalFrames) {
    printf(" Window Slides");
   frame[base].frameNo);
  }
}
```



Sample Input

Enter Message : Hello

Window Size : 3

Enter frame No to lose : 0.

Sample Output:

Sender window : [1:h] [2:e] [3:l]

Frames

Sender → Frame 1, Data : h

Sender → Frame 2, Data : e

Sender → Frame 3, Data : l

Receiver

Frame 1, Data : h

Rece ← ACK 1

Frame 2, Data : e

Receiver ← ACK 2

Frame 3, Data : l

Receiver ← Ack 3.

Window Slides forward to frame 4.

Sender window : [4:l][5:0].

Receiver ← ACK 4.

Reciver ← ACK 5.

All the frames are transmitted Successfully.

Result:

The Blinding window for transmitting the frames are learned and executed Successfully.